

FINANCIAL FRAUD DETECTION MODEL WITH DASHBOARD

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I would like to express my sincere gratitude to Amdox Technologies for providing me with the opportunity to work as a Data Science & Analytics Intern. This internship helped me gain practical knowledge in Machine Learning, Data Analysis, Data Visualization, Database Management, and Fraud Detection Systems.

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I would also like to thank my institute, teachers, and family members for their continuous support and motivation during the internship period.

ABSTRACT

Financial fraud has become one of the major challenges faced by banks and digital payment systems. The increasing number of online transactions creates opportunities for fraudulent activities that may result in significant financial losses.

The objective of this project is to develop a Machine Learning based Financial Fraud Detection System capable of identifying fraudulent transactions from a large-scale financial dataset. The project utilizes the PaySim Financial Transaction Dataset, performs data preprocessing, feature engineering, exploratory data analysis, machine learning model training, model evaluation, database integration, and dashboard visualization.

Three machine learning algorithms including Logistic Regression, Decision Tree, and Random Forest were implemented and compared. Random Forest achieved the highest performance with an F1 Score of 98.18% and Accuracy of 99.995%.

A Streamlit-based interactive dashboard was also developed to visualize transaction statistics and fraud patterns.

Keywords: Machine Learning, Fraud Detection, Random Forest, Data Analytics, Streamlit Dashboard, Financial Transactions.

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INTRODUCTION

Financial fraud detection is a critical area in the banking and financial sector. Fraudulent transactions can result in substantial monetary losses and damage customer trust. Traditional rule-based systems often fail to detect sophisticated fraud patterns.

Machine Learning techniques can automatically learn transaction behavior and identify suspicious activities with higher accuracy. This project focuses on building an intelligent fraud detection model using supervised learning techniques.

PROBLEM STATEMENT

The financial sector processes millions of transactions daily. Identifying fraudulent transactions among legitimate transactions is challenging because fraud cases represent only a very small percentage of the overall data.

The problem is to develop a machine learning system that accurately detects fraudulent transactions while minimizing false alarms.

OBJECTIVES

The primary objectives of this project are:

- Analyze financial transaction data.
- Perform data preprocessing and cleaning.
- Create new features using feature engineering.
- Train multiple machine learning models.
- Compare model performance.
- Identify important fraud indicators.
- Store processed data in a database.
- Build an interactive fraud monitoring dashboard.

DATASET OVERVIEW

Dataset Used: PaySim Financial Transaction Dataset

Dataset Source: Kaggle PaySim Financial Transaction Dataset.

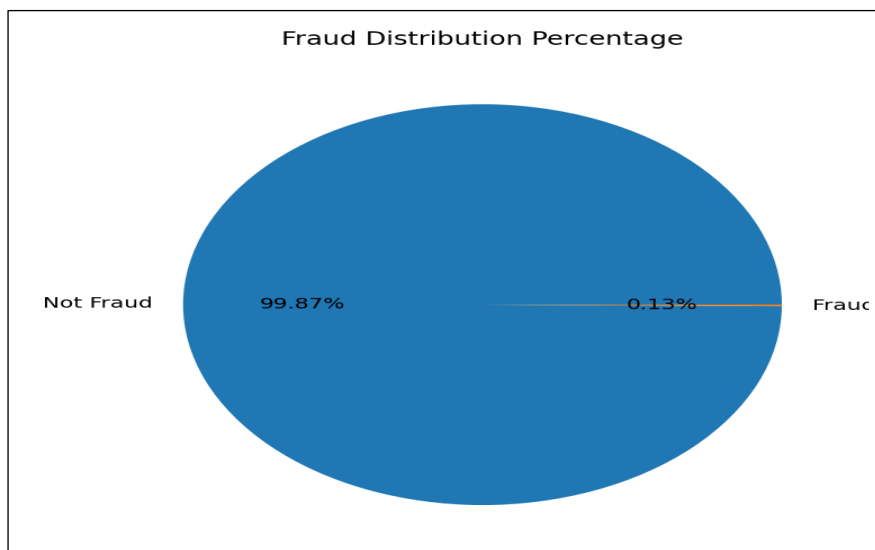
Dataset Statistics:

- Total Transactions: 6,362,620
- Fraudulent Transactions: 8,213
- Non-Fraudulent Transactions: 6,354,407
- Fraud Percentage: 0.129%

Features:

- step
- type
- amount
- oldbalanceOrg
- newbalanceOrig
- oldbalanceDest
- newbalanceDest
- isFraud
- isFlaggedFraud

Figure 1: Fraud Distribution Pie Chart



DATA PREPROCESSING

The preprocessing stage included:

- Missing value analysis
- Duplicate record removal
- Label encoding of categorical variables
- Data type verification

Results:

- Missing Values: 0
- Duplicate Records: 0

The dataset was clean and ready for model training.

FEATURE ENGINEERING

Three additional features were created:

Sender Balance Difference

$\text{org_balance_diff} = \text{oldbalanceOrg} - \text{newbalanceOrig}$

Receiver Balance Difference

$\text{dest_balance_diff} = \text{newbalanceDest} - \text{oldbalanceDest}$

Amount Ratio

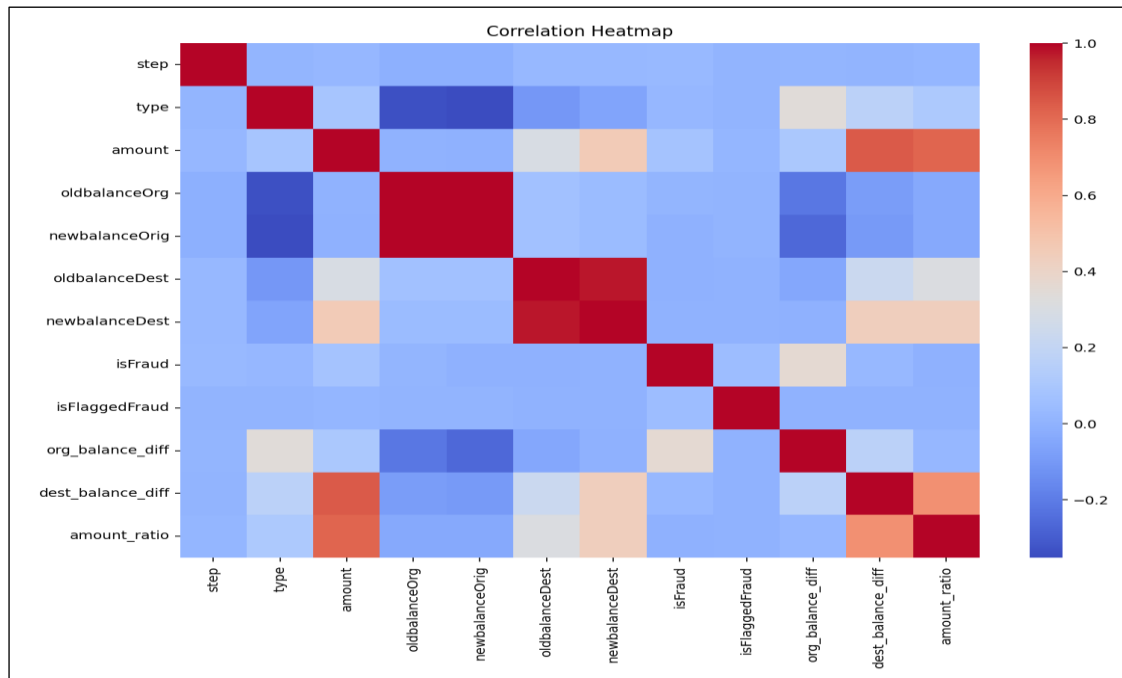
$\text{amount_ratio} = \text{amount} / (\text{oldbalanceOrg} + 1)$

These features significantly improved model performance.

EXPLORATORY DATA ANALYSIS

EDA was performed to understand transaction behavior and fraud patterns.

Figure 2: Correlation Heatmap



Observations:

- Strong correlations exist between balance-related features.
- Fraud is associated with abnormal balance changes.
- Transaction amount plays an important role in fraud detection.

The correlation heatmap shows the relationships between different transaction features in the dataset. Strong positive correlations are observed between account balance variables such as oldbalanceOrg and newbalanceOrig and oldbalanceDest and newbalanceDest. The transaction amount also has a significant relationship with engineered features like amount_ratio and dest_balance_diff. These relationships indicate that balance changes and transaction amounts play an important role in identifying fraudulent transactions. Overall, the heatmap helps in understanding feature dependencies and selecting important variables for building an effective fraud detection model.

MACHINE LEARNING MODELS

Three supervised learning algorithms were implemented:

Logistic Regression

Used as a baseline classification model.

Decision Tree

Provides rule-based fraud detection and interpretability.

Random Forest

An ensemble model that combines multiple decision trees to improve accuracy and reduce overfitting.

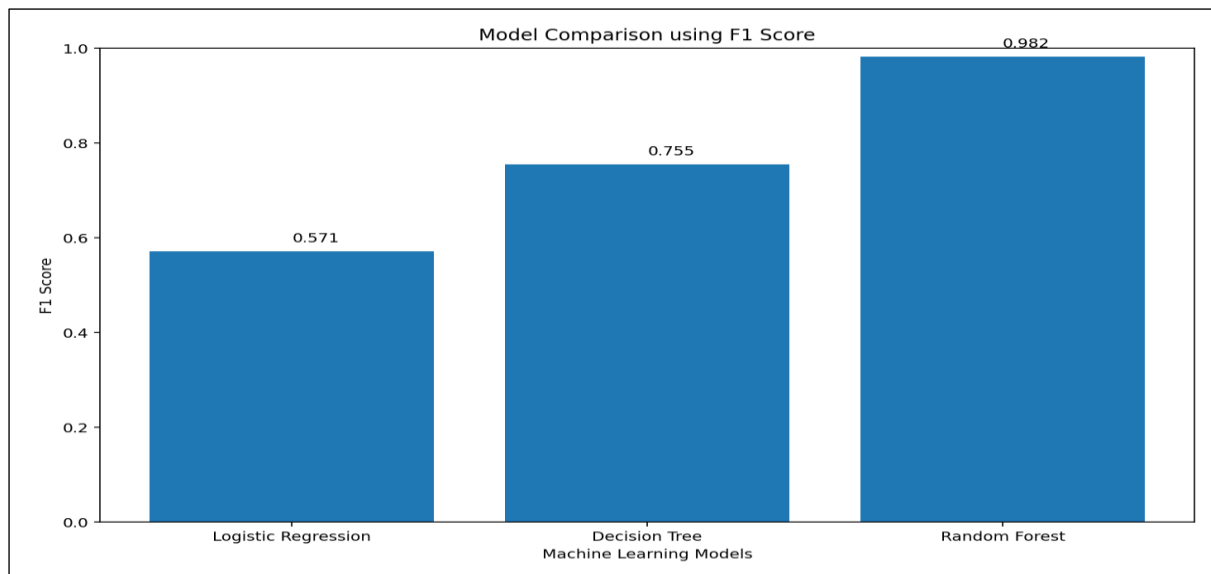
MODEL EVALUATION

Performance Metrics:

- Accuracy
- Precision
- Recall
- F1 Score

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	99.91%	85.71%	42.86%	57.14%
Decision Tree	99.94%	80.00%	71.43%	75.47%
Random Forest	99.995%	100.00%	96.43%	98.18%

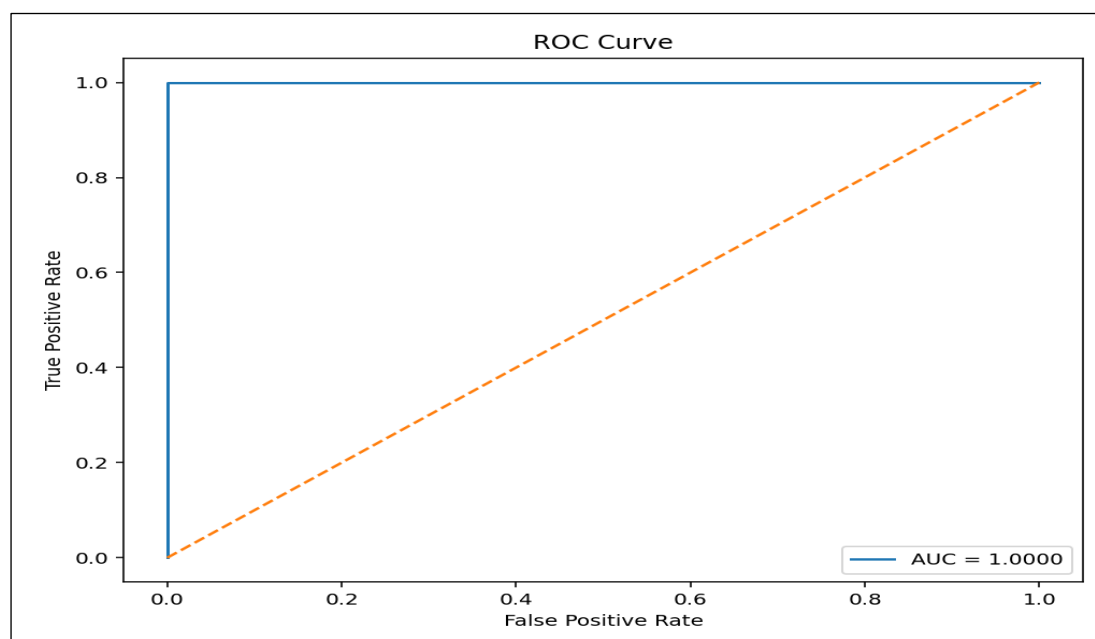
Figure 3: Model Comparison Chart



The model comparison chart presents the F1 Scores of the three machine learning models used for fraud detection: Logistic Regression, Decision Tree, and Random Forest. Among all models, Random Forest achieved the highest F1 Score of 0.982, indicating superior performance in identifying fraudulent transactions while minimizing classification errors. Decision Tree achieved an F1 Score of 0.755, whereas Logistic Regression obtained 0.571. Based on the comparison, Random Forest was selected as the final model due to its excellent balance between precision and recall, making it the most effective algorithm for financial fraud detection in this project.

ROC Curve

Figure 4: ROC Curve

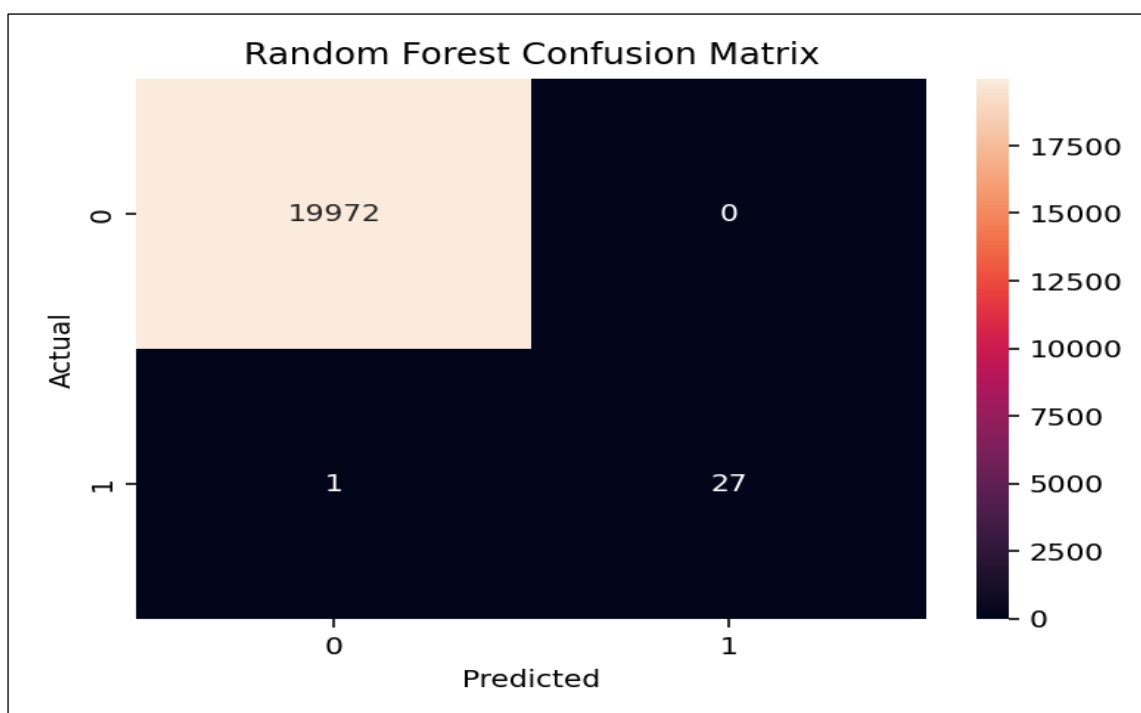


The ROC (Receiver Operating Characteristic) Curve evaluates the classification performance of the Random Forest model by comparing the True Positive Rate (Sensitivity) against the False Positive Rate. The curve is positioned near the top-left corner of the graph, indicating excellent discrimination between fraudulent and non-fraudulent transactions. The model achieved an AUC (Area Under the Curve) score of 1.0000, which represents near-perfect classification performance. This result demonstrates that the Random Forest model is highly effective in detecting fraudulent transactions while minimizing false positives, making it the most suitable model for this fraud detection system.

Random Forest achieved an AUC score of 1.0000 indicating excellent classification capability.

Confusion Matrix

Figure 5: Random Forest Confusion Matrix



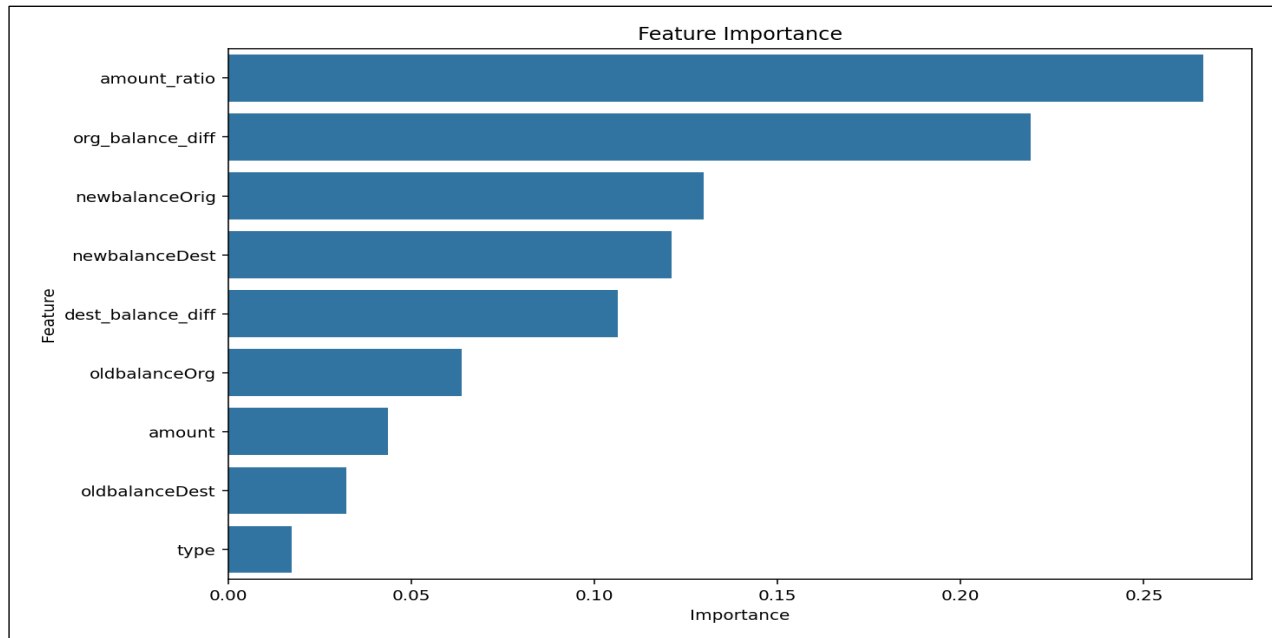
Results:

- True Negatives: 19,972
- False Negatives: 1
- False Positives: 0
- True Positives: 27

The confusion matrix evaluates the classification performance of the Random Forest model by comparing actual and predicted transaction classes. The model correctly classified 19,972 legitimate transactions (True Negatives) and 27 fraudulent transactions (True Positives). It produced 0 False Positives, meaning no legitimate transaction was incorrectly classified as fraud, and only 1 False Negative, where a fraudulent transaction was missed. These results demonstrate the excellent performance of the Random Forest model, achieving very high accuracy and reliability in detecting fraudulent transactions while minimizing classification errors.

Feature importance analysis identified the most influential variables used by the Random Forest model.

Figure 6: Feature Importance Bar Chart



Top Features:

1. amount_ratio
2. org_balance_diff
3. newbalanceOrig
4. newbalanceDest
5. dest_balance_diff

These variables contribute significantly to fraud prediction.

DATABASE INTEGRATION

SQLite database was used to store processed transaction records and fraud prediction results. Database integration enables efficient data management, historical record tracking, and future deployment of the fraud detection system.

Database Name:

fraud_detection.db

Benefits:

- Efficient storage
- Historical transaction tracking
- Future deployment support

FRAUD PREDICTION SYSTEM

The trained Random Forest model was saved using Joblib.

Model File:

fraud_model.pkl

The saved model can be loaded to classify new transactions without retraining.

INTERACTIVE DASHBOARD

A Streamlit-based dashboard was developed to visualize transaction data and fraud statistics.

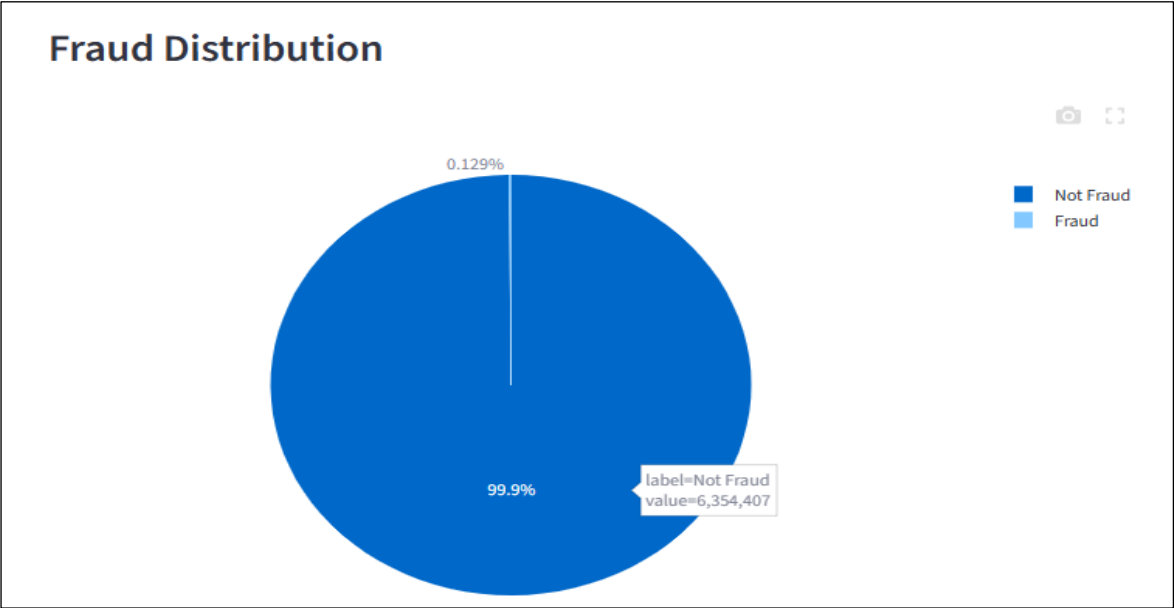
Dashboard Features:

- KPI Cards
- Fraud Distribution Analysis
- Transaction Type Analysis
- Fraud by Transaction Type
- Transaction Amount Distribution
- Top Fraud Transactions Table

Figure 7: Dashboard Overview with KPI Cards

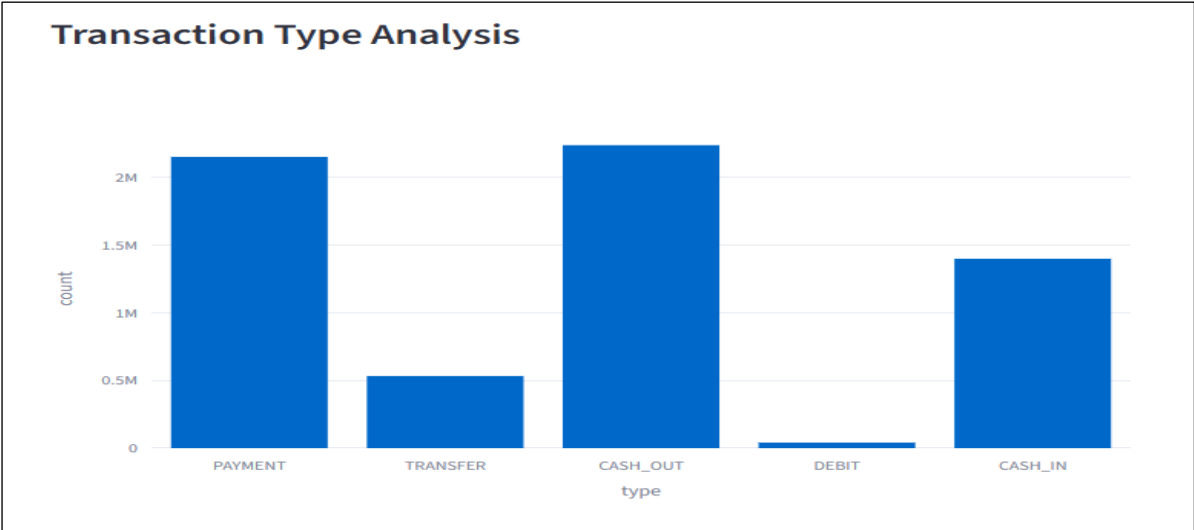
Total Transactions	Fraud Transactions	Fraud Rate %
6,362,620	8213	0.129

Figure 8: Fraud Distribution Dashboard



The Fraud Distribution Dashboard visualizes the proportion of fraudulent and non-fraudulent transactions in the dataset. The chart shows that 99.87% of transactions are legitimate, while only 0.13% are fraudulent, indicating a highly imbalanced dataset. This imbalance highlights the challenge of fraud detection, as fraudulent transactions represent a very small fraction of the overall data. The dashboard provides a clear overview of transaction distribution and emphasizes the importance of using robust machine learning techniques to accurately identify rare fraud cases.

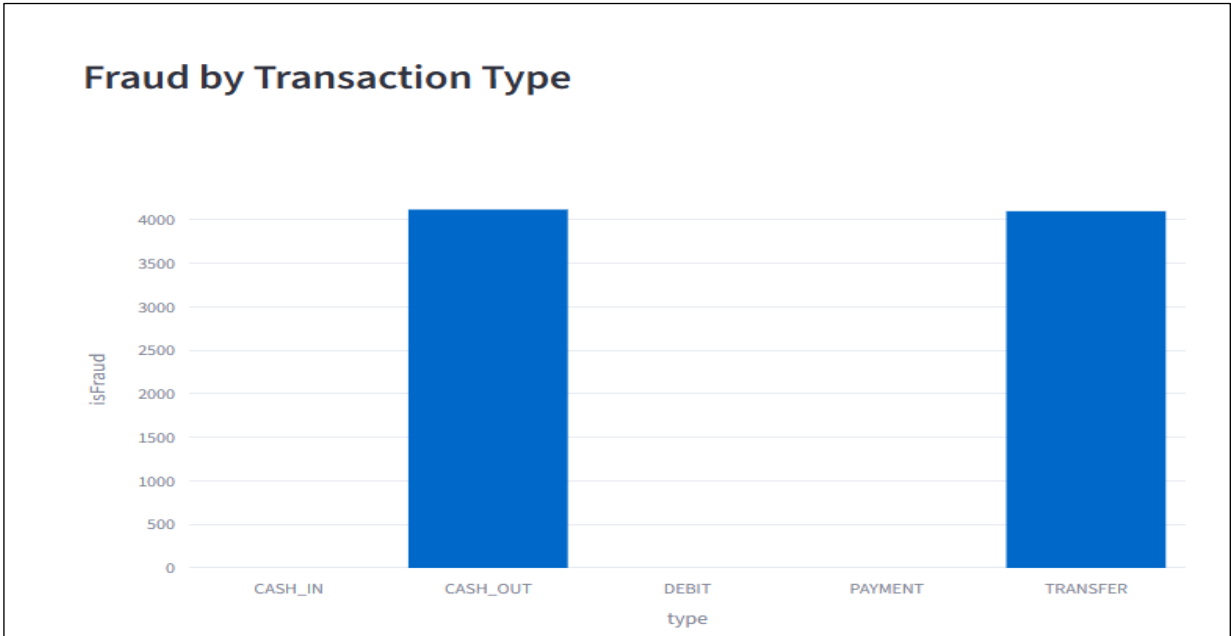
Figure 9: Transaction Type Analysis Dashboard



The Fraud Distribution Dashboard visualizes the proportion of fraudulent and non-fraudulent transactions in the dataset. The chart shows that 99.87% of transactions are legitimate, while only 0.13%

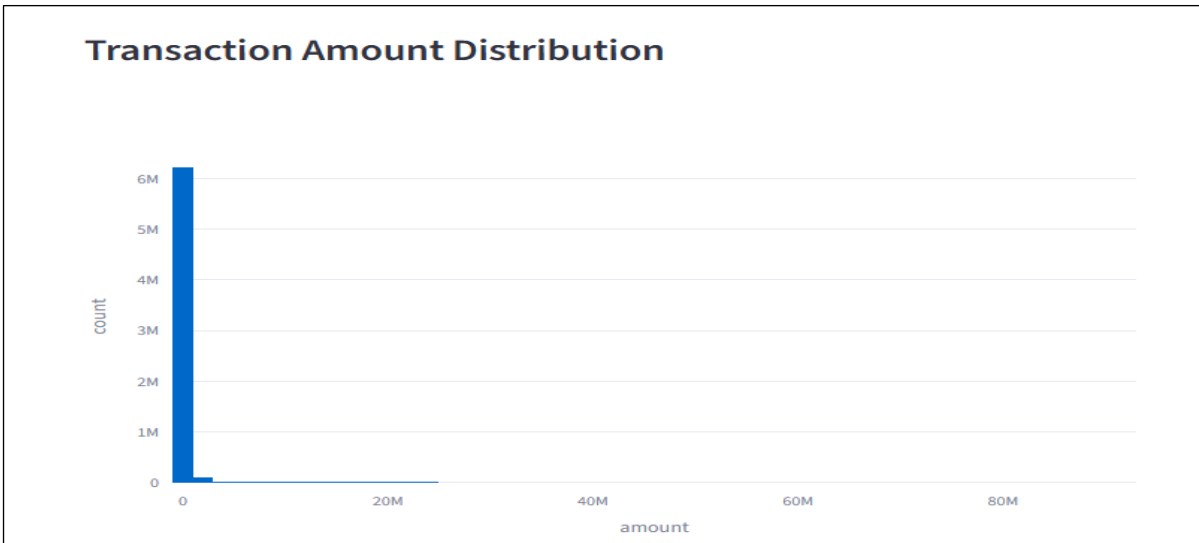
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Figure 10: Fraud by Transaction Type Dashboard



The Fraud by Transaction Type Dashboard shows the distribution of fraudulent transactions across different transaction categories. The chart indicates that fraudulent activities occur primarily in CASH_OUT and TRANSFER transactions, with both transaction types recording a similar number of fraud cases. In contrast, PAYMENT, DEBIT, and CASH_IN transactions show little to no fraudulent activity. This observation suggests that fraudsters mainly exploit transfer and cash withdrawal operations to conduct unauthorized transactions. These insights are valuable for financial institutions, as they help focus fraud monitoring and prevention efforts on the most vulnerable transaction types.

Figure 11: Transaction Amount Distribution Dashboard



The Transaction Amount Distribution Dashboard illustrates the frequency distribution of transaction amounts in the dataset. The histogram shows that the majority of transactions involve relatively small amounts, which are concentrated near the lower end of the distribution. As transaction amounts increase, the number of transactions decreases significantly, resulting in a highly right-skewed distribution. A small number of transactions involve extremely large amounts, which may represent unusual or high-risk activities. This analysis helps in understanding transaction behavior and highlights the importance of monitoring high-value transactions, as they may have a greater likelihood of being associated with fraudulent activities

Figure 12: Top Fraud Transactions Dashboard

	step	type	amount	nameOrig	oldbalanceOrig	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud	
	2	1	TRANSFER	181	C1305486145	181	0	C553264065	0	0	1	0
	3	1	CASH_OUT	181	C840083671	181	0	C38997010	21182	0	1	0
	251	1	TRANSFER	2806	C1420196421	2806	0	C972765878	0	0	1	0
	252	1	CASH_OUT	2806	C2101527076	2806	0	C1007251739	26202	0	1	0
	680	1	TRANSFER	20128	C137533655	20128	0	C1848415041	0	0	1	0
	681	1	CASH_OUT	20128	C1118430673	20128	0	C339924917	6268	12145.85	1	0
	724	1	CASH_OUT	416001.33	C749981943	0	0	C667346055	102	9291619.62	1	0
	969	1	TRANSFER	1277212.77	C1334405552	1277212.77	0	C431687661	0	0	1	0
	970	1	CASH_OUT	1277212.77	C467632528	1277212.77	0	C716083600	0	2444985.19	1	0
	1115	1	TRANSFER	35063.63	C1364127192	35063.63	0	C1136419747	0	0	1	0
	1116	1	CASH_OUT	35063.63	C1635772897	35063.63	0	C1983025922	31140	7550.03	1	0
	1869	1	TRANSFER	25071.46	C669700766	25071.46	0	C1384210339	0	0	1	0
	1870	1	CASH_OUT	25071.46	C1275464847	25071.46	0	C1364913072	9083.76	34155.22	1	0
	1911	1	CASH_OUT	132842.64	C13692003	4499.08	0	C297927961	0	132842.64	1	0
	2301	1	TRANSFER	235238.66	C1872047468	235238.66	0	C116289363	0	0	1	0

The Top Fraud Transactions Dashboard displays a sample of transactions that have been identified as fraudulent by the dataset. The table provides detailed information about each suspicious transaction, including transaction type, amount, sender and receiver account details, account balances before and after the transaction, and fraud status.

From the dashboard, it can be observed that most fraudulent transactions belong to the TRANSFER and CASH_OUT categories. Many of these transactions involve unusual balance changes, where the sender's account balance decreases significantly while the receiver's balance remains unchanged or shows abnormal patterns. Such inconsistencies are strong indicators of fraudulent behavior.

This dashboard enables users to inspect suspicious transactions in detail, helping financial institutions understand fraud patterns, investigate high-risk activities, and support decision-making for fraud prevention and risk management.

Benefits:

- Real-time monitoring
- Fraud trend visualization
- Interactive analysis

PROJECT OUTCOMES

- Successfully detected fraudulent transactions.
- Achieved 99.995% accuracy using Random Forest.
- Developed an interactive Streamlit dashboard.
- Implemented database integration using SQLite.
- Generated actionable insights from transaction data.

CONCLUSION

A Financial Fraud Detection Model was successfully developed using Machine Learning techniques. Multiple models were evaluated, and Random Forest achieved the best performance with an Accuracy of 99.995% and an F1 Score of 98.18%.

The project successfully integrated data preprocessing, feature engineering, machine learning, database management, and dashboard visualization into a complete fraud detection solution.

FUTURE SCOPE

Future improvements may include:

- XGBoost and LightGBM implementation
- Deep Learning based fraud detection
- Real-time fraud monitoring systems
- Cloud deployment using AWS or Azure
- Mobile application integration
- SMS and Email alert systems

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